Physics Unit 2 · Practice Quiz 3 (Challenge)

Static and kinetic friction · inclined planes · universal gravitation · circular motion · Newton's laws

Name: Date: Period:

Instructions. Part A: 20 multiple-choice questions, 2 points each (40 points). Part B: 4 free-response questions, 10 points each (40 points). Total 80 points. Suggested time: 50 minutes.

Reference. $g = 9.81\ m/s^2$ · $G = 6.67 \times 10^{-11}\ N{\cdot}m^2/kg^2$ · $M_E = 5.97 \times 10^{24}\ kg$ · $R_E = 6.38 \times 10^{6}\ m$. Give answers to 3 significant figures with units. Show your work on the free-response questions. A calculator may be used (degree mode).

Part A · Multiple choice (2 points each)

1. A 40 kg box rests on a level floor ($\mu_s = 0.50$). You pull it with 200 N at 30° above the horizontal. Does it start to move?

A No: the horizontal pull (173 N) is less than $\mu_s mg$ (196 N)

B Yes: the pull (200 N) is more than $\mu_s mg$ (196 N)

C Yes: the horizontal pull (173 N) is more than $f_{s,max}$ (146 N)

D No: the pull (200 N) is less than $f_{s,max}$ (146 N)

2. A 1.5 kg book is pressed against a vertical wall by a horizontal push ($\mu_s = 0.40$). What is the smallest push that keeps the book from sliding down?

A 36.8 N **B** 3.75 N

C 14.7 N **D** 5.89 N

3. A box sits on the flat bed of a truck that accelerates forward at 2.5 m/s². What is the smallest μ_s that keeps the box from sliding?

A 2.5 **B** 3.9

C 0.25 **D** 0.75

4. A 20 kg sled is pulled across level snow by a rope at 30° above the horizontal with a 80 N force ($\mu_k = 0.15$). What is its acceleration?

A 1.99 m/s² **B** 2.29 m/s²

C 3.46 m/s² **D** 2.53 m/s²

5. You slowly increase your push on a box until it just starts to slide, then keep pushing with that same force. If $\mu_s = 0.50$ and $\mu_k = 0.35$, what is the box's acceleration?

A 1.47 m/s² **B** 3.43 m/s²

C 4.91 m/s² **D** 0 m/s²

6. A hockey puck leaves a stick at 12 m/s and slides 60 m across level ice before it stops. What is μ_k?

A 0.24 **B** 0.061

C 0.12 **D** 1.2

7. A block is given a push and slides UP a 30° incline at 6.0 m/s ($\mu_k = 0.20$). How far up the slope does it slide before stopping?

A 5.61 m **B** 5.45 m
C 3.67 m **D** 2.73 m

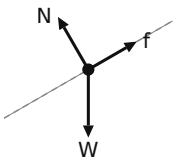
8. What force, parallel to the slope, pushes a 20 kg crate UP a 30° ramp at constant speed? ($\mu_k = 0.25$)

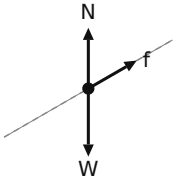
A 42.5 N **B** 141 N
C 98.1 N **D** 55.6 N

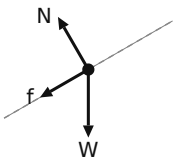
9. A block is placed at rest on a 30° incline. $\mu_s = 0.50$. What happens?

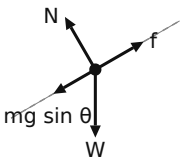
A It slides, because every block on a slope slides
B It slides, because tan 30° = 0.58 is greater than μ_s
C It stays at rest, because tan 30° = 0.58 is greater than μ_s
D It stays at rest, because sin 30° = 0.50 is less than μ_s

10. A block rests on a rough incline without sliding. Which is the correct free-body diagram? (The dashed line is the slope.)

A


B


C


D


11. Two objects attract each other with a gravitational force F. One mass is tripled and the distance is doubled. What is the new force?

A 3F/2 **B** 4F/3
C 12F **D** 3F/4

12. A satellite orbits 400 km above Earth's surface. What is its orbital speed? ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.38 \times 10^6$ m)

A 10,800 m/s **B** 31,600 m/s
C 58.7 m/s **D** 7,660 m/s

13. A satellite orbits 400 km above Earth's surface. What is its orbital period? ($M_E = 5.97 \times 10^{24}$ kg, $R_E = 6.38 \times 10^6$ m)

A 46.3 min **B** 5.47 min
C 92.6 min **D** 29.5 min

14. At what distance from Earth's CENTER is g one quarter of its surface value?

A 4 Earth radii
B 2 Earth radii (one Earth radius above the surface)
C ¼ of an Earth radius
D 16 Earth radii

15. At what angle should a curve of radius 100 m be banked so cars can take it at 20 m/s with no friction?

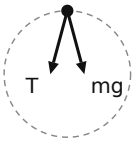
A 22.2° **B** 1.2°
C 67.8° **D** 24.1°

16. A 0.5 kg ball on a 0.8 m string is swung in a vertical circle. At the BOTTOM its speed is 4.0 m/s. What is the tension in the string?

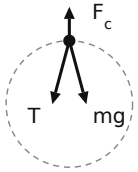
A 5.09 N **B** 14.9 N
C 10.0 N **D** 4.91 N

17. A ball on a string moves in a vertical circle. Which is the correct free-body diagram at the TOP of the circle?

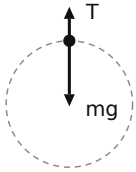
A



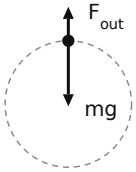
B



C



D



18. A carousel horse is 5.0 m from the center and goes around once every 8.0 s. What are its speed and centripetal acceleration?

A $v = 1.96$ m/s, $a = 0.771$ m/s² **B** $v = 3.93$ m/s, $a = 0.785$ m/s²
C $v = 0.625$ m/s, $a = 0.0781$ m/s² **D** $v = 3.93$ m/s, $a = 3.08$ m/s²

19. A 4.0 kg block on a table ($\mu_k = 0.25$) is tied over a pulley to a 2.0 kg hanging mass. What is the acceleration?

A 4.91 m/s² **B** 3.27 m/s²
C 2.45 m/s² **D** 1.64 m/s²

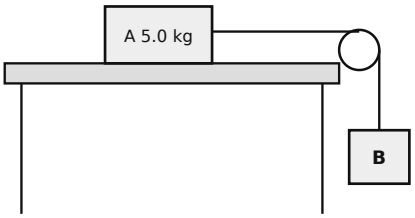
20. A 10 kg sign hangs at rest from two ropes, each at 30° above the horizontal, one on each side. What is the tension in each rope?

A 196 N **B** 56.6 N
C 98.1 N **D** 49.1 N

Part B · Free response (10 points each)

21. Connected objects with friction (10 points)

Block A (5.0 kg) sits on a table with $\mu_s = 0.45$ and $\mu_k = 0.30$. It is tied over a light, frictionless pulley to a hanging block B.



(a) What is the largest mass of B that will NOT start A moving? **[3]**

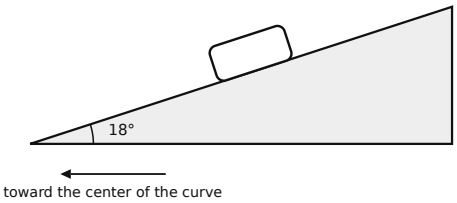
(b) B has a mass of 3.0 kg. Find the acceleration of the blocks. **[3]**

(c) Find the tension in the string, and check it using block A. **[2]**

(d) Explain why the tension is less than the weight of B. **[2]**

22. Banked curve (10 points)

A highway curve of radius 120 m is banked at 18°.



(a) Draw the free-body diagram of a car taking the curve at the design speed (no friction needed). **[2]**

(b) Find the design speed. **[3]**

(c) A car takes the curve at 25 m/s. Is the inward part of the normal force enough? Which way must friction act on the car (up or down the slope)? **[3]**

(d) On an icy day (no friction) a car takes the curve at 25 m/s. Describe its motion. **[2]**

23. Geostationary orbit (10 points)

A geostationary satellite stays above the same point on the equator, so its period is 24.0 h.

(a) Starting from $GMm/r^2 = mv^2/r$ with $v = 2\pi r/T$, show that $r^3 = GMT^2/(4\pi^2)$, and find the orbital radius r. **[4]**

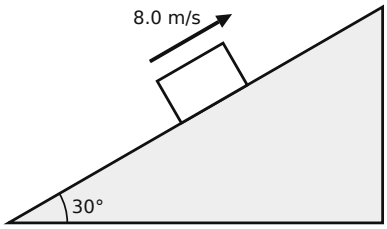
(b) How high above Earth's surface is the satellite? **[1]**

(c) Find its orbital speed. **[2]**

(d) Find g at that orbit. If gravity still acts on the satellite, why would an astronaut inside feel weightless? **[3]**

24. Up and back down an incline (10 points)

A block is launched UP a 30° incline at 8.0 m/s. $\mu_k = 0.20$ and $\mu_s = 0.40$.



(a) Find the block's deceleration as it slides up. **[2]**

(b) How far up the slope does it slide? **[2]**

(c) When it stops, does it stay there or slide back down? Show your reasoning. **[2]**

(d) Find its acceleration on the way back down. **[2]**

(e) How fast is it moving when it gets back to the starting point? Why is this less than 8.0 m/s? **[2]**